

## MINICOY, India

WMO: 433690

Lat: 8.281N

Lon: 73.054E

Elev: 7

StdP: 14.69

Time Zone: 5.50 (IND)

Period: 94-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |       |      |      |       |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-------|------|------|-------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |       |      | 99%  |       |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR    | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)   | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 74.0       | 75.1 | 68.4                          | 104.5 | 80.8 | 69.6 | 108.8 | 80.6 | 11.9                  | 82.9 | 10.6 | 84.0 | 1.2                   | 270  | 0.375 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 9.3                    | 92.1            | 82.0 | 91.4 | 81.6 | 90.6 | 81.3 | 83.4                | 89.7 | 82.8 | 89.1 | 82.4 | 88.6 | 5.1                  | 320  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 81.7                            | 164.7 | 87.6 | 81.0 | 161.1 | 87.1 | 80.6 | 159.0 | 86.8 | 47.4          | 90.2 | 46.7 | 89.6 | 46.2 | 88.9 | 86.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)   |    |    |
| (4) 17.3          | 13.9 | 11.5 | 68.8                       | 93.9 | 4.6                | 2.4 | 65.5                                               | 95.6 | 62.8       | 97.0 | 60.2       | 98.4 | 56.8       | 100.1 |    |    |
| (5)               |      |      | 66.8                       | 84.8 | 3.7                | 0.7 | 64.2                                               | 85.3 | 62.0       | 85.8 | 59.9       | 86.2 | 57.2       | 86.7  |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 83.5  | 82.2  | 82.7  | 84.6  | 86.0  | 85.7  | 83.8  | 83.1  | 82.9  | 82.8  | 82.4  | 82.4  | (6)  |      |
| (7)  |                                                                                  | DBStd       | 2.24  | 1.40  | 1.60  | 1.57  | 1.78  | 2.18  | 2.25  | 2.25  | 2.03  | 1.99  | 1.85  | 1.64  | 1.63 | (7)  |
| (8)  |                                                                                  | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                                  | HDD65       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (9)  |
| (10) |                                                                                  | CDD50       | 12218 | 997   | 915   | 1074  | 1079  | 1106  | 1013  | 1027  | 1027  | 986   | 1018  | 973   | 1003 | (10) |
| (11) |                                                                                  | CDD65       | 6743  | 532   | 495   | 609   | 629   | 641   | 563   | 562   | 562   | 536   | 553   | 523   | 538  | (11) |
| (12) |                                                                                  | CDH74       | 79223 | 5717  | 5514  | 7540  | 8320  | 8389  | 6782  | 6522  | 6484  | 6045  | 6187  | 5777  | 5946 | (12) |
| (13) |                                                                                  | CDH80       | 29883 | 1840  | 1926  | 3231  | 4050  | 3980  | 2606  | 2274  | 2228  | 1961  | 2017  | 1847  | 1923 | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.3   | 2.7   | 3.0   | 2.9   | 3.1   | 4.5   | 7.5   | 7.6   | 6.3   | 5.1   | 3.8   | 2.3   | 2.6  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 62.4  | 1.5   | 0.8   | 1.1   | 2.0   | 6.1   | 11.5  | 10.2  | 7.8   | 5.9   | 6.7   | 6.1   | 2.8  | (15) |
| (16) |                                                                                  | PrecMax     | 75.9  | 5.0   | 5.2   | 9.3   | 7.2   | 14.1  | 27.0  | 16.6  | 14.6  | 14.3  | 12.1  | 16.8  | 8.8  | (16) |
| (17) |                                                                                  | PrecMin     | 36.9  | 0.1   | 0.0   | 0.0   | 0.0   | 1.9   | 4.3   | 2.2   | 1.2   | 0.7   | 1.7   | 1.5   | 0.0  | (17) |
| (18) |                                                                                  | PrecStd     | 9.8   | 1.3   | 1.2   | 2.2   | 1.7   | 3.1   | 4.1   | 3.5   | 3.3   | 3.2   | 2.7   | 3.5   | 2.8  | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 89.1  | 89.9  | 91.8  | 93.1  | 93.3  | 91.4  | 90.0  | 89.9  | 90.0  | 89.7  | 89.6  | (19) |      |
| (20) |                                                                                  |             | MCWB  | 78.8  | 79.4  | 80.8  | 82.5  | 82.5  | 82.8  | 81.8  | 81.4  | 80.6  | 80.6  | 80.5  | 80.1 | (20) |
| (21) |                                                                                  | 2%          | DB    | 88.2  | 89.0  | 91.0  | 92.2  | 92.2  | 90.0  | 89.1  | 89.1  | 88.9  | 89.1  | 88.7  | 88.6 | (21) |
| (22) |                                                                                  |             | MCWB  | 78.4  | 79.0  | 80.5  | 82.1  | 82.2  | 82.2  | 81.5  | 81.1  | 80.4  | 80.4  | 80.0  | 79.3 | (22) |
| (23) |                                                                                  | 5%          | DB    | 87.6  | 88.3  | 90.3  | 91.4  | 91.4  | 89.3  | 88.3  | 88.2  | 88.0  | 88.2  | 87.9  | 87.8 | (23) |
| (24) |                                                                                  |             | MCWB  | 78.1  | 78.7  | 80.2  | 81.7  | 82.0  | 81.9  | 81.1  | 80.8  | 80.2  | 80.2  | 79.5  | 78.7 | (24) |
| (25) |                                                                                  | 10%         | DB    | 86.9  | 87.6  | 89.4  | 90.6  | 90.3  | 88.3  | 87.4  | 87.3  | 87.1  | 87.3  | 87.1  | 86.9 | (25) |
| (26) |                                                                                  |             | MCWB  | 77.9  | 78.4  | 79.9  | 81.3  | 81.8  | 81.5  | 80.8  | 80.5  | 80.0  | 79.9  | 79.3  | 78.4 | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 80.8  | 81.0  | 82.4  | 83.9  | 84.1  | 83.7  | 82.8  | 82.6  | 82.3  | 82.7  | 82.1  | 82.6 | (27) |
| (28) |                                                                                  |             | MCDWB | 86.9  | 87.8  | 90.0  | 90.8  | 90.7  | 89.5  | 88.3  | 88.3  | 88.0  | 88.0  | 87.6  | 87.3 | (28) |
| (29) |                                                                                  | 2%          | WB    | 80.0  | 80.1  | 81.5  | 83.1  | 83.4  | 82.8  | 82.0  | 81.8  | 81.5  | 81.7  | 81.0  | 80.9 | (29) |
| (30) |                                                                                  |             | MCDWB | 86.1  | 87.0  | 89.1  | 90.1  | 89.9  | 88.6  | 87.6  | 87.5  | 87.1  | 87.1  | 86.6  | 86.2 | (30) |
| (31) |                                                                                  | 5%          | WB    | 79.4  | 79.7  | 80.8  | 82.4  | 82.8  | 82.4  | 81.6  | 81.3  | 80.8  | 81.0  | 80.5  | 80.1 | (31) |
| (32) |                                                                                  |             | MCDWB | 85.4  | 86.4  | 88.4  | 89.6  | 89.2  | 88.1  | 86.9  | 87.0  | 86.5  | 86.4  | 86.0  | 85.3 | (32) |
| (33) |                                                                                  | 10%         | WB    | 78.8  | 79.1  | 80.2  | 81.8  | 82.3  | 81.8  | 81.0  | 80.7  | 80.2  | 80.4  | 79.9  | 79.4 | (33) |
| (34) |                                                                                  |             | MCDWB | 84.7  | 85.6  | 87.4  | 88.7  | 88.6  | 87.1  | 86.1  | 86.3  | 85.6  | 85.6  | 85.0  | 84.6 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 10.4  | 10.7  | 10.2  | 9.3   | 8.7   | 7.9   | 7.5   | 8.0   | 8.1   | 8.7   | 9.1   | 9.7  | (35) |
| (36) |                                                                                  |             | MCDBR | 10.8  | 11.2  | 10.5  | 9.6   | 9.3   | 8.9   | 8.5   | 8.8   | 9.0   | 9.7   | 10.2  | 10.3 | (36) |
| (37) |                                                                                  |             | MCWBR | 4.2   | 4.2   | 3.9   | 3.9   | 4.0   | 4.4   | 4.3   | 4.3   | 4.2   | 4.4   | 4.3   | 4.3  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 9.7   | 10.3  | 10.1  | 9.3   | 8.8   | 8.6   | 8.1   | 8.5   | 8.5   | 8.9   | 9.2   | 9.1  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.1   | 4.3   | 4.1   | 4.3   | 4.3   | 4.6   | 4.5   | 4.5   | 4.4   | 4.6   | 4.3   | 4.3  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.475 | 0.483 | 0.528 | 0.529 | 0.510 | 0.526 | 0.550 | 0.505 | 0.460 | 0.460 | 0.468 | 0.479 | (40) |      |
| (41) |                                                                                  | taud        | 2.238 | 2.213 | 2.103 | 2.127 | 2.205 | 2.151 | 2.055 | 2.213 | 2.367 | 2.360 | 2.315 | 2.256 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 266   | 268   | 257   | 253   | 252   | 245   | 240   | 256   | 271   | 271   | 266   | 262   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 45    | 47    | 53    | 51    | 46    | 48    | 53    | 46    | 40    | 40    | 41    | 43    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1822  | 1979  | 2057  | 1970  | 1694  | 1489  | 1582  | 1713  | 1771  | 1697  | 1557  | 1648  | (44) |      |
| (45) |                                                                                  | RadStd      | 63    | 81    | 104   | 113   | 187   | 130   | 156   | 139   | 177   | 122   | 109   | 99    | (45) |      |

## Historical Trends

|      |                        | Heating |         | Cooling |         |         | Degree-Days |         |         |         |         |      |
|------|------------------------|---------|---------|---------|---------|---------|-------------|---------|---------|---------|---------|------|
|      |                        | 99% DB  | 99% DP  | 1% DB   | 1% WB   | 1% DP   | HDD50       | HDD65   | CDD50   | CDD65   |         |      |
| (46) | Station Only           | (d) N/A | (e) N/A | (f) N/A | (g) N/A | (h) N/A | (i) N/A     | (j) N/A | (k) N/A | (l) N/A | (m) N/A | (46) |
| (47) | Regional (0 neighbors) | N/A     | N/A     | N/A     | N/A     | N/A     | N/A         | N/A     | N/A     | N/A     | N/A     | (47) |

Nomenclature: See separate page